MILITARY SPECIFICATION

RESISTOR NETWORK, FIXED, FILM (EXTERNALLY TRIMMABLE)

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements not normally covered by the drawing and provides quality assurance provisions for the fabrication of a Trimmable Resistor Network for use in the BLU-92/B Mine, Antipersonnel and Mine, Antitank: BLU-91/B.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-R-39032	-	Resistors, Preparation For Delivery Of
MIL-R-83401	-	Resistor Networks, Fixed Film, General Specification For

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-202	-	Test Methods For Electronic And Electrical Component Parts
MIL-STD-883	-	Test Methods And Procedures For Microelectronics

FSC: 1345

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-1188 - Commercial Packaging Of Supplies And Equipment
MIL-STD-1285 - Marking Of Electrical And Electronic Parts

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9328687 - Network, Resistor, Fixed and Ajustable

3. REQUIREMENTS

3.1 General. The manufacturer of the resistor network in compliance with this specification shall have production and test facilities and a quality and reliability assurance program adequate to assure successful compliance with the provisions of this specification. Only resistor networks which are inspected for and meet all of the requirements specified herein shall be marked and delivered.

3.2 Detail specification. The complete requirements for the network shall be in accordance with Drawing (DWG) 9328687 (detailed drawing), as specified in this detailed specification, and MIL-R-83401.

3.3 Design and construction. The resistor network shall consist of a ceramic substrate with conductors, insulating and resistance patterns applied by thick film (TF) techniques in the order necessary to meet the requirements of this specification. The resistance patterns for the trimmable resistors (R2, R3, and R5) shall be on the top side of the substrate and positioned for YAG laser trimming as specified in detailed drawing. The dc resistance between terminals shall be in accordance with the resistance table and network schematic in detailed drawing.

3.3.1 Solderability. Terminal leads shall be free of foreign material, flash marks, and other irregularities and conform to the requirements of MIL-STD-202, Test Method 208.

3.3.2 Enclosure. The resistor network shall be encapsulated sufficiently to withstand the environmental tests specified. The encapsulating material shall be compatible with YAG laser trimming.

3.3.3 Physical dimensions. The physical dimensions of the resistor network package shall comply with all requirements in detailed drawing. Package dimension inspection shall be performed in accordance with MIL-STD-883, Method 2016.

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3.4 Electrical characteristics.

3.4.1 Network characteristics. When resistor networks are subjected to the condition or test specified in Table I, the percent change in resistance in each resistor element shall not exceed the value specified in Table I. The examination and test requirement method shall be in accordance with MIL-R-83401 for characteristic M resistor network unless otherwise specified herein.

3.4.2 Maximum voltage. The maximum continuous working voltage applied to the resistor elements shall be 20 volts d.c. or 20 volts rms (root-mean-square, sine-wave).

3.4.3 Resistor

3.4.3.1 Fixed resistors. The nominal resistance and tolerance for each resistor element shall be as specified in DWG 9328687.

3.4.3.2 Trimmable resistors R2, R3, and R5. The trimmable resistors shall not be trimmed by the resistor network manufacturer; however, their initial resistance prior to user laser trimming may be less than the minimum specified for their trimmable resistance range. (See detailed drawing for the trimmable resistance range). The initial resistance value shall be the nominal resistance used in the dc resistance tests. The trimmable resistor shall be designed so that any resistance within its trimmable resistance range shall be obtainable with a single straight cut made with a Yag Laser set with a spot size of 2 mils, Q-switched at 3000 pulses per second, and with power of 1 watt. Windows shall expose all trimmable material and the substrate around this material as specified in detailed drawing.

3.5 Device marking.

3.5.1 Marking. Each resistor network shall be marked in accordance with Method 1 of MIL-STD-1285 as follows:

a. Part number - 9328687X (X = Revision letter).

b. Supplier's code symbol.

c. Manufacturer's date code, the first two digits shall be the last two digits of the number of the year. The third and fourth digits shall indicate the calendar week of the year.

d. Pin No. 1 identifier, see DWG 9328687.

3.5.2 Resistance to solvents. The markings on resistor networks shall be legible after being subjected to the resistance to solvents test of MIL-STD-202, Method 215.

3.5.3 Data requirements. Data shall be recorded on a go/no-go basis and retained for all test called out by this specification. (See 6.4).

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3.6 First article inspection. This specification contains provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Workmanship. The trimmable resistive network shall be fabricated and finished in a thorough, workmanlike manner consistent with the best practices of the industry. It shall be free of corrosion products, and other foreign matter. The cleaning method used shall not be injurious to the resistive network nor shall the parts be contaminated by the cleaning agents. All markings and stampings shall be free of harmful cleaning and soldering agents that may be injurious to the thick film (TF). All markings and stampings, shall be displayed and visibly defined. The TF shall be manufactured and/or processed in such a manner as to be uniform in quality and free from defects that will affect life, configuration and serviceability.

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TABLE I. Network Characteristics

Test or Condition	MIL-R-83401 Requirement-Method Paragraphs 1/	Value	Units
Resistance-temperature characteristic	3.13 - 4.6.8	± 250	Parts/ million/°C
Maximum ambient temperature at rated wattage	3.5, 3.6 - Fig. 1	+ 70	Degrees Celsius
Maximum ambient temperature at zero power derating	3.5, 3.6 - Fig. 1	+125	Degrees Celsius
Thermal shock	3.7 - 4.6.3	± 0.5	
Power conditioning	3.8 - 4.6.4	± 0.5	
Low-temperature operation	3.14 - 4.6.9	± 0.5	Maximum per-
Short time overload	3.15 - 4.6.10	± 0.5	cent change
Terminal strength	3.16 - 4.6.11	± 0.25	in resistance
Resistance to soldering heat	3.19 - 4.6.14	± 0.25	(0.01 ohm additional
Moisture resistance	3.20 - 4.6.15	± 0.5	allowed for
Shock (specified pulse)	3.21 - 4.6.16	± 0.25	measurement
High-frequency vibration	3.22 -4.6.17	± 0.25	error)
Life	3.23 - 4.6.18	± 2.0	
High temperature exposure	3.2.4 - 4.6.19	± 1.0	
Low-temperature storage	3.25 - 4.6.20	± 0.5	
D.C. resistance 2/	3.9 - 4.6.5	2/	
*Insulation resistance	3.18 - 4.6.13	20,000	Megohms

*These characteristics or conditions vary from Code Symbol M in MIL-R-83401.

1/ Headings of requirement and method paragraph of MIL-R-83401 (General Specification) are the same as listed above for test or condition. The headings are the prime references. Paragraph numbers are given to facilitate locating references.

2/ The nominal resistance and resistance tolerance shall be as specified in detailed drawing (see 3.3, 3.4.3.2).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-R-83401 shall apply and are hereby made a part of this specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of sixty (60) trimmable resistor networks.

4.3.2 Inspections to be performed. First article inspection shall be performed by the supplier, after award of contract and prior to production, at a location acceptable to the Government (see 6.3). It shall be performed on sample units which have been produced with equipment and procedures normally used in production. See Table II specified herein.

4.3.3 Rejection. If any resistor network fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a resistor network in the sample, to comply with any of the stated requirements.

4.3.4 Requalification. The resistor network shall be subjected to another first article if:

a. The contractor's manufacturing techniques or process controls which affect form, fit, function or internal geometry of the device are revised by either contractor or one of his sub-contractors supplying parts or material.

b. The contractor relocates his manufacturing facilities.

c. The contractor uses a new sub-contractor for material, parts or assembly.

d. A change in material used for resistor network process or assembly.

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TABLE II. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-R-63528(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Resistor Network, Fixed, Film (Externally Trimmable)				9328687
					NEXT HIGHER ASSEMBLY
					--
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Group I Non-destructive	60	-		MIL-R-83401
	Thermal shock			3.4.1	4.6.3
	Power conditioning			3.4.1	4.6.4
	DC resistance			3.4.1	4.6.5
	Group IA Non-destructive				
	Physical dimensions			3.3.3	MIL-STD-883, Method 2016
	Visual and mechanical examinations			3.3.2,3.5 3.2, 3.3 3.7	4.5.1 herein
	Group II	10	-		
	Solderability			3.3.1	MIL-STD-202, Method 208
	Resistance to solvents			3.5.2	MIL-STD-202, Method 215
	Group III	10	-		MIL-R-83401
	Resistance-temperature characteristic			3.4.1	4.6.8, Note 1,Sheet 2
	Low temperature operation			3.4.1	4.6.9
	Short time overload			3.4.1	4.6.10
	Terminal strength			3.4.1	4.6.11

NOTES: All sixty (60) sample units shall be subjected to Group I tests and IA examinations and provided there are zero rejects then divided as shown for groups II thru VII inclusive and subjected to the tests for their particular group. All groups tests and examinations are to be conducted in the order shown.

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TABLE II. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-R-63528(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Resistor Network, Fixed, Film (Externally Trimmable)				9328687
					NEXT HIGHER ASSEMBLY --
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Group IV	10	-		MIL-R-83401
	Insulation resistance			3.4.1	4.6.13
	Resistance to soldering heat			3.4.1	4.6.14
	Moisture resistance			3.4.1	4.6.15
	Group V	10	-		MIL-R-83401
	Shock (specified pulse)			3.4.1	4.6.16
	Vibration, high frequency			3.4.1	4.6.17
	Group VI	10	-		
	Life			3.4.1	4.6.18
	Group VII	10	-		
	Low temperature storage			3.4.1	4.6.20
	High temperature exposure			3.4.1	4.6.19

NOTES:

1. Follow the sequence called out by MIL-R-83401, Table XI, column TITLE "Qualification Inspection".

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-R-83401.

4.4.2 Examination. Inspection of product for delivery shall consist of groups A and B. (See 4.4.2.1 through 4.4.2.6).

4.4.2.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in 4.4.2.4, and shall be made on the same set of sample units, in the order shown.

a. Sampling plan. Resistor networks that are out of resistance tolerance or which experience a change in resistance greater than that permitted for the tests of subgroup 1 shall be removed from the lot. Lots having more than 10 percent total rejects, due to exceeding the specified resistance change limit, shall not be furnished on the contract. Sampling and inspection for subgroup 1 and 2 shall be in accordance with MIL-STD-105 for general inspection level II.

4.4.2.2 Group B inspection. Group B inspection shall consist of the tests specified in 4.4.2.5, in the order shown. They shall be performed on sample units that have been subjected to and have passed the Group A inspection.

a. Sampling plan. Sampling and inspection shall be in accordance with MIL-STD-105 for inspection level S-4. Samples units which have passed all the Group B inspections may be delivered on the contract provided they are within resistance tolerance and meet requirements for visual and mechanical inspection.

4.4.2.3 Group C inspection. Group C inspection shall consist of the tests specified in 4.4.2.6, in the order shown. They shall be performed periodically, as specified in 4.4.2.6, on sample units selected from lots that have passed groups A and B inspection.

a. Sampling plan. The sample size for each subgroup in 4.4.2.6 shall be 10 units. Sample units which have been subjected to group C inspection shall not be delivered on the contract. If rejects exceeds those allowed in 4.4.2.6, supplier shall perform failure analysis and furnish copy to technical agency listed in 6.4.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-R-63528(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Resistor Network, Fixed, Film (Externally Trimmable)	1 OF 1	9328687
			NEXT HIGHER ASSEMBLY --

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Group A, Subgroup 1 (non-destructive tests)				
	Thermal shock		0.40%	3.4.1	MIL-R-83401, 4.6.3
	Power conditioning		0.40%	3.4.1	MIL-R-83401, 4.6.4
	DC resistance		0.40%	3.4.1	MIL-R-83401, 4.6.5
	Group A, Subgroup 2 (non-destructive)				
	Physical dimensions		0.65%	3.3.3	MIL-STD-883, Method 2016
	Visual and mechanical examinations	1/	0.65%	3.2,3.3, 3.3.2, 3.5,3.7	4.5.1, herein
	Solderability	10	0-1	3.3.1	MIL-STD-202, Method 208

NOTES: 1/ Body: cracks or holes which would cause probable failure; Leads: leads broken, crushed or nicked which would cause probable failure in use, lead spacing incorrect; Markings: incorrect, illegible.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-R-63528(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Resistor Network, Fixed, Film (Externally Trimmable)				9328687
					NEXT HIGHER ASSEMBLY --
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Group B (non-destructive)				
	Resistance-temperature characteristic		2.5%	3.4.1	MIL-R-83401, 4.6.8
	Insulation resistance		2.5%	3.4.1	MIL-R-83401, 4.6.13
	Short-time overload		2.5%	3.4.1	MIL-R-83401, 4.6.10

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-R-63528(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Resistor Network, Fixed, Film (Externally Trimmable)	1 OF 1	9328687
			NEXT HIGHER ASSEMBLY: --

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Group C, Monthly 2/	10	1-2		
	Low-temperature operation			3.4.1	MIL-R-83401, 4.6.9
	Short-time overload			3.4.1	MIL-R-83401, 4.6.10
	Terminal strength			3.4.1	MIL-R-83401, 4.6.11
	Group C, Subgroup 1, Quarterly 2/	10	1-2		
	Insulation resistance			3.4.1	MIL-R-83401, 4.6.13
	Resistance to soldering heat			3.4.1	MIL-R-83401, 4.6.14
	Moisture resistance			3.4.1	MIL-R-83401, 4.6.15
	Subgroup 2 2/	10	1-2		
	Life			3.4.1	MIL-R-83401, 4.6.18
	Group C, Subgroup 2, Semiannually 2/	10	1-2		
	Shock (specified pulse)			3.4.1	MIL-R-83401, 4.6.16
	Vibration, high frequency			3.4.1	MIL-R-83401, 4.6.17
	Subgroup 2 2/	10	0-1		
	Solderability			3.3.1	MIL-STD-202, Method 208
	Resistance to solvents			3.5.2	MIL-STD-202, Method 215
	Subgroup 3 2/	10	1-2		
	High temperature exposure			3.4.1	MIL-R-83401, 4.6.19
	Low temperature storage			3.4.1	MIL-R-83401, 4.6.20

NOTES: 1/ The ten (10) samples per subgroup: samples shall be subjected to the tests in each subgroup in order shown. 2/ DESTRUCTIVE TESTS: Sample units shall not be delivered on the contract.

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection forms.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and test prescribed herein is described in the 'Paragraph Reference/Inspection Method' column of the First Article and Quality Conformance forms. Design responsibility for all inspection equipment is assigned to the contractor, who shall submit for approval inspection equipment design in accordance with data item description DI-E-1118 or DI-R-1714, with the appropriate addendum to each, and with the terms of the contract. See 4.1.1 "Test equipment and inspection facilities" of MIL-R-83401.

4.5 Methods of inspections. The methods of inspection and testing are given by reference in the First Article and Quality Conformance forms and 4.5.1 herein.

4.5.1 Visual and mechanical examination. Resistor networks shall be examined to verify that the materials, design, construction, markings, and workmanship are in accordance with the applicable requirements (See 3.2, 3.3, 3.3.2, 3.3.3, 3.5 and 3.7).

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing

5.2 Packaging. The resistor network shall be packaged in accordance with level B of MIL-R-39032.

5.3 Packing. The resistor network shall be packed in accordance with level B of MIL-R-39032.

5.4 Marking. The packed components shall be marked in accordance with the commercial requirements of MIL-STD-1188.

5.5 Shipping. When resistor networks from more than one lot are shipped together, each lot shall be kept separate, and the divisions between lots clearly indicated to prevent mixing of lots in transit.

6. NOTES

6.1 Intended use. The networks covered by this specification are intended for single-use, non-repairable military devices.

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6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Provisions of submission of first article samples.

c. Quantity required and delivery schedules.

d. Certificates of conformance for each lot or shipment of product.

6.3 Submission of inspection equipment designs for approval. (See 4.4.4). Submit equipment designs as required to Commander: ARRADCOM, ATTN: DRDAR-QAT-I, Dover, New Jersey 07801.

6.4 Distribution of data (Lot traveller(s) inspection and test results). Distribution of data shall include the following: Commander, ARRADCOM, ATTN: DRDAR-QAT-M, Dover, New Jersey 07801.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.6 Changes from previous issue. Due to the extensiveness of the changes, asterisks are not used in this revision to identify changes with respect to the previous issue.

Custodian:
Army-AR

Preparing Activity:
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